

Machine Learning Assessment of Geosite Accessibility in Morocco: National-Scale Application

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Abstract

Physical accessibility governs whether a geologically significant site can be turned into a geotourism or educational asset. We assess accessibility for 1,662 Moroccan geosites (733 from the original citation-traceable inventory, 206 added by a second domain-expert labeler, and 722 added by a third 2026-08-28 labeling batch, all cross-checked against each other and against a data-quality audit of the master catalog's region assignment before use), spanning all twelve administrative regions with labeled sites, using terrain and infrastructure variables under a spatially rigorous validation protocol designed to avoid the geographic data leakage that commonly inflates reported accuracy in similar spatial-prediction studies. At national scale, a six-feature tree-ensemble baseline reaches 76.7% (Difficult) and 71.3% (Easy) accuracy against majority-class baselines of 78.9% and 59.6% respectively: at this larger sample size the Difficult classifier is now significantly *below* its trivial baseline ($p = 0.039$), while the Easy classifier is now significantly *above* its trivial baseline ($p < 0.001$) – both directions newly resolved from an earlier, smaller sample where neither reached significance. Two feature additions and a full model-family comparison were used to search for a better Difficult model specifically: a geological-domain feature and a tourism-infrastructure feature each deliver a statistically confirmed accuracy gain over the six-feature baseline for both classifiers ($p \leq 0.018$ throughout), and, more consequentially, a Gaussian Process classifier significantly outperforms the deployed tree ensemble on Difficult once given the same feature additions ($p = 0.020$) – a reversal of this line of work's earlier finding that no alternative model family displaces the tree ensemble. The final deployed models reported throughout are therefore heterogeneous by design, chosen by evidence rather than fixed in advance: a Gaussian Process classifier on the infrastructure-augmented feature set for Difficult (80.8% accuracy) and the tree ensemble on the same feature set for Easy (73.5%), the tree ensemble remaining the confirmed better choice for that target under every feature set tested. A direct three-class model reaches 53.4% accuracy, statistically indistinguishable from the two-binary-classifier combination it was compared against ($p = 0.946$) – both approaches are equally valid, interchangeable framings of the same result. A companion geosite-location favorability model (AUC 0.956) answers a related but distinct question – where currently uncatalogued geosites are likely to be found – each result illustrated with a dedicated, professionally rendered map. Split-conformal prediction sets, validated on genuinely held-out data, show the two deployed models diverge in how often they can commit to a single confident class: roughly seven in ten sites for Difficult, roughly one in two for Easy, with an honest "could be either" for the rest – a more actionable statement of what each model actually knows than a single accuracy figure. A full per-region comparative breakdown of performance, robustness, and limitations is the dedicated subject of a companion paper; we close here by situating this national-scale work relative to two published accessibility studies.

1 Introduction

Morocco's national geoheritage inventory includes geological formations of real scientific and educational value, but many are only useful for outreach, teaching, or tourism if visitors can actually reach them. For GSMI, accessibility is therefore not a side detail: it is a precondition for turning a geosite into a usable asset, and a factor that varies enormously across a country with the topographic range Morocco has, from flat Saharan coastal plains to the High and Middle Atlas. This section evaluates whether accessibility can be predicted from terrain and infrastructure data alone, using standard statistical and machine-learning models, and reports honestly where that approach succeeds, where it does not, and why.

Figure 1 situates the underlying dataset geographically before any modeling: all 1,662 labeled geosites, plotted by their true, expert-assigned accessibility class. Two patterns are visible immediately and recur throughout this section: labeled geosites cluster heavily in the Middle Atlas and Fés-Meknès corridor, where geoheritage documentation has historically been densest, and Difficult sites concentrate along the Rif and Middle Atlas mountain fronts, while the southern regions and Oriental are comparatively sparsely labeled – a coverage pattern that directly shapes which regional results below are statistically trustworthy.

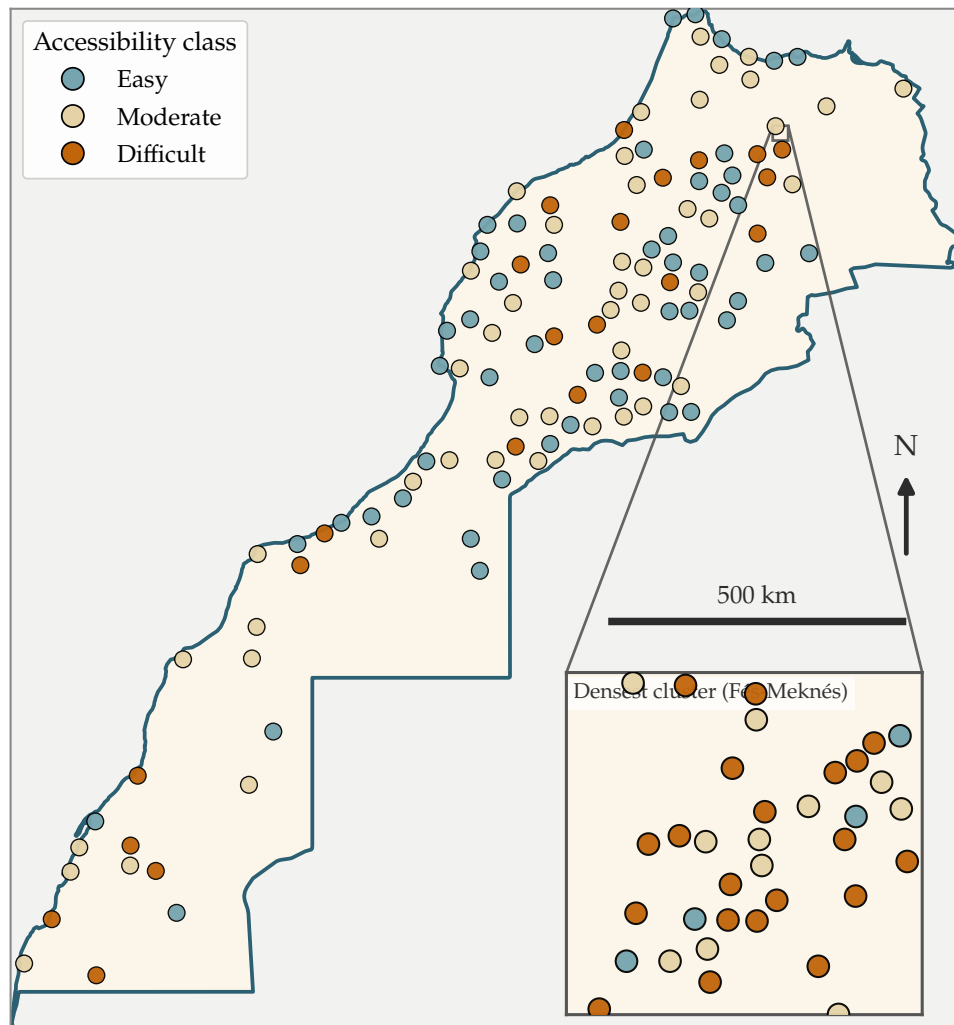


Figure 1. Labeled geosites, by true accessibility class – a spatially thinned sample is shown for legibility, with the inset magnifying the single densest cluster; all 1,662 sites are used throughout the analysis (Table 1).

2 Study Area and Dataset

The analysis covers 1,662 labeled geosites across Morocco’s administrative regions (Table 1), each assigned one of three accessibility classes – *Easy*, *Moderate*, or *Difficult* – based on how a visitor would realistically reach the site (road type, remaining walking distance, and terrain along the final approach). Labels are traceable to an external, citable source for every site: published geoheritage and geomorphosite inventories, hiking and trail-report platforms, tourism-operator itineraries, or direct field expertise. Sites for which no independently verifiable source could be found were left unlabeled rather than assigned by inference. 733 of the 1,662 labels come from the original citation-traceable inventory, 206 were added by a second domain-expert labeler, and 722 were added by a third labeling batch (2026-08-28) covering nearly the full 1,667-site catalog. All batches were audited before use – typo and non-standard region-name correction, geometric region reassignment for coordinate/boundary mismatches, and manual resolution of every coordinate-duplicate label conflict against the underlying source material. Cross-checking the third batch against the first two surfaced ten disagreements (1% of the 940 overlapping sites), three of them the same same-coordinate cluster already documented in an earlier audit; the already-audited label was kept in every case. The third batch also independently reproduced four more same-coordinate label conflicts of the same kind, at previously-unlabeled sites – those four sites were excluded rather than arbitrarily resolved. Reconciling the third batch’s labels against the catalog’s Region field additionally surfaced a data-quality bug – Morocco’s twelve official regions were represented by 21 distinct spelling/accent variants in the master catalog, only 8 of 289 affected sites having ever been labeled before, so the bug was invisible until this batch labeled the rest of them. It was corrected via point-in-polygon spatial join against the official region boundary file, which also corrected a handful of sites genuinely mis-assigned to the wrong region (not just misspelled) – this is why regional totals below cannot be obtained by simply adding the previously-published N=939 table to the new batch’s raw counts. Per-source expert-confidence weighting, used in an earlier iteration of this work, has been retired: every labeler’s contribution is treated on equal footing, not as a lower-confidence addition.

Table 1. Dataset composition: labeled geosites by region and accessibility class (N=1,662).

Region	Difficult	Easy	Moderate	Total
Fés-Meknés	192	193	152	537
Béni Mellal-Khénifra	54	91	181	326
Tanger-Tétouan-Al Hoceima	39	160	113	312
Drâa-Tafilalet	27	86	57	170
Souss-Massa	11	37	48	96
Marrakech-Safi	8	42	19	69
Rabat-Salé-Kénitra	4	21	19	44
Guelmim-Oued Noun	2	23	13	38
Eddakhla-Oued Eddahab	8	3	23	34
Laayoune-Sakia El Hamra	3	7	8	18
Casablanca-Settat	2	6	6	14
Oriental	0	2	2	4
Total	350	671	641	1,662

Every region grew substantially in this update – Fés-Meknés from 337 to 537 labeled sites, Tanger-Tétouan-Al Hoceima from 128 to 312, Béni Mellal-Khénifra from 174 to 326 – and Oriental, previously unlabeled, now carries 4 sites (2 Easy, 2 Moderate, still none Difficult). Several regions still carry very few labeled sites in one or more classes (Casablanca-Settat, $n = 14$; Oriental, $n = 4$;

Laayoune-Sakia El Hamra, $n = 18$). This uneven coverage is a direct, identifiable constraint on what regional models can be trained and trusted, and is treated explicitly in Section 6 rather than left implicit.

3 Methodology

3.1 Predictor Variables

Six terrain and infrastructure variables, the common baseline feature set for every model in this study, were computed for each site from open geospatial datasets (Table 2), grouped here by what each group of variables physically represents. Section 4.1 introduces two further feature-set additions (a geological-domain encoding, and tourism-infrastructure indicators) tested on top of this baseline; the models ultimately deployed and mapped in this paper use the infrastructure-augmented set, not the baseline alone.

Table 2. Predictor variables used in all models, grouped by physical meaning.

Variable	Description	Source
Terrain factors – what the ground itself is like at the site		
Slope (degrees)	Local terrain slope at the site	Copernicus DEM
Ruggedness	Terrain ruggedness (elevation variability in the surrounding window)	Copernicus DEM
Elevation (m)	Site elevation above sea level	Copernicus DEM
Infrastructure & land cover – what stands between a visitor and the site		
Dist. to highway (m)	Distance to the nearest paved road segment	OSM road network
Dist. to settlement (m)	Distance to the nearest village/settlement point	OSM place points
LULC friction	Land-cover-based movement resistance coefficient	ESA WorldCover

3.2 Labeling Protocol

Each site’s accessibility class reflects how a visitor would reach it in practice: road type and condition up to the nearest drivable point, and the terrain difficulty of any remaining walk. Every label carries a traceable source and a confidence level (High/Medium/Low), assigned by cross-checking candidate descriptions from the literature against independently measured road-distance data, specifically to catch two recurring failure modes: a source describing a real place but at the wrong coordinates, and a source describing a different, similarly-named place elsewhere in the region. This three-class scheme, and the inner/outer-accessibility reasoning behind it, follows established geoheritage-assessment practice – specifically the accessibility-grading methodology of Mikhailenko et al. [4] and the broader geoheritage assessment framework of Reynard and Brilha [5].

3.3 Validation Strategy

Two geosites a few hundred meters apart are, for practical purposes, the same access problem: same road, same approach, same difficulty. If such near-duplicate pairs are split across training and test sets by an ordinary random split, a model can score well simply by copying the label of a site it has effectively already seen, without learning anything transferable about terrain or infrastructure – illustrated schematically in Figure 2. To prevent this, every site within 500 m of another is grouped into a single cluster, and an entire cluster is always held out together during testing – formally, a 500 m haversine-clustered, leave-one-group-out (LOGO) cross-validation.

Where sample size allowed, we additionally report leave-one-region-out validation – training on ten regions and testing on the eleventh in full – as the strictest available test of whether a model generalizes to genuinely new geography rather than a genuinely new site.

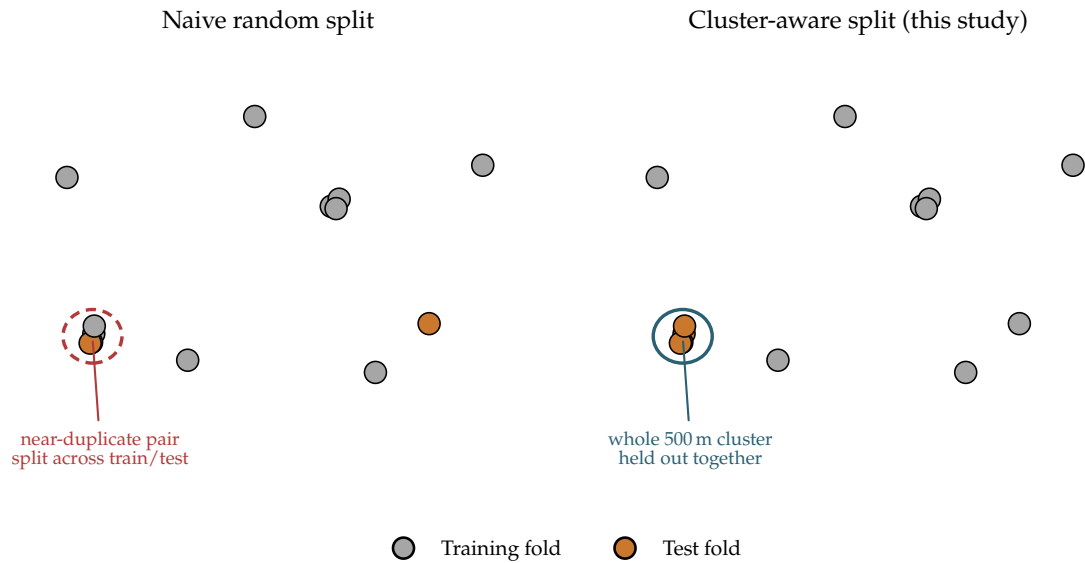


Figure 2. Schematic illustration (synthetic points, not real data) of the two validation strategies. Left: a naive random split can place near-duplicate sites on both sides of the train/test boundary. Right: this study’s cluster-aware split always keeps an entire 500 m cluster on one side.

A third protocol, spatial block cross-validation (partitioning the study area into fixed-size geographic tiles and holding whole tiles out, rather than clusters of near-duplicate points), is used specifically for the geosite-location favorability model (Section 4.4): that model’s presence/background design produces far denser, more spatially continuous point coverage than the accessibility labels, where block CV is the standard, better-suited protocol for testing whether the model has learned genuine geological structure rather than fine-grained spatial autocorrelation. Table 3 summarizes which protocol backs which result in this paper; the three are not interchangeable substitutes for each other but answer three different robustness questions.

Table 3. Cross-validation protocols used in this paper, what each one tests, and where it is applied.

Protocol	What it tests	Used for
LOGO-cluster CV	Near-duplicate sites never split across train/test	All binary and 3-class results (Section 4)
Leave-region-out	Generalization to a region with zero training data	Companion regional paper
Spatial block CV	Robustness to spatial autocorrelation, on dense point coverage	Favorability model only (Section 4.4)

3.4 Models Compared

The working default model in the rest of this section is a four-member tree-ensemble vote (Random Forest, XGBoost, Gradient Boosting, and LightGBM), soft-voted and trained with class-balance-weighted samples. Three additional, structurally unrelated model families – a linear baseline, k -nearest-neighbors, and a Gaussian Process classifier – were tested under the

identical 500 m LOGO-cluster protocol (Gaussian Process under a disclosed exception: 10-fold group cross-validation rather than full LOGO, since its $O(n^3)$ per-fold cost makes 1,391 individual folds impractical) as a baseline-choice check, per Table 4.

Table 4. Model-family comparison, $N=1,662$, six-feature baseline, same 500 m LOGO-cluster protocol (Gaussian Process: 10-fold group CV, see text). Best k for k -NN chosen by LOGO-cluster accuracy among $k \in \{5, 10, 15, 25\}$.

Model	Difficult acc. (%)	Easy acc. (%)
Tree ensemble	76.7	71.3
Logistic Regression	68.9	66.1
k -NN ($k = 10$ Diff., $k = 25$ Easy)	79.7	68.7
Gaussian Process	79.2	69.9

The picture that held at the smaller sample size does not hold at this one. A paired McNemar test against the tree ensemble finds Difficult: k -NN $\chi^2 = 10.73$, $p = 0.0011$; GP $\chi^2 = 7.20$, $p = 0.0073$ – both alternatives now *significantly beat* the tree ensemble, not just numerically edge it out. Easy inverts in the opposite direction: k -NN $\chi^2 = 12.87$, $p = 0.0003$, the tree ensemble now *significantly beating* k -NN; GP is not significant ($\chi^2 = 1.85$, $p = 0.173$, carrying the same 10-fold-vs-full-LOGO caveat as before). The linear baseline trails clearly on both targets throughout and is not considered further. This is a genuine, evidence-driven reversal of this line of work’s earlier conclusion that no alternative displaces the tree ensemble – at this sample size, for the Difficult target specifically, it does. Section 4.2 extends this comparison to the feature-set additions introduced in Section 4 and resolves the final model choice used throughout the rest of this paper; it is deliberately not resolved here, since the better-performing feature sets found later change which comparison actually matters. (A separate, five-member variant that additionally includes CatBoost was used only for an internal cluster-radius sensitivity check, reported in the companion regional-comparison paper – it was not used to produce any of the headline results in Table 6.)

3.5 Three-Way Decision Rule

A direct single three-class (Easy/Moderate/Difficult) model was tried during early exploration on the original $N=733$ dataset and abandoned there: it underperformed and was harder to interpret than a two-step binary approach. On the $N=939$ audited dataset this comparison was rerun and reversed (a direct 3-class model reaching 56.2%, ahead of two-binary combinations at 54.4–54.5%, borderline significant). At the current $N=1,662$, re-run end to end again rather than assumed settled, the comparison reverses back: a direct 3-class model now reaches 53.4% LOGO-cluster accuracy (majority-class baseline 40.4%), against 53.9% for the exclusion rule described below and 53.4% for a probability-based ordinal combination that additionally enforces internal consistency between the two binaries (detailed below) – all three approaches now sit within half a point of each other. A paired McNemar test between the direct 3-class model and the ordinal binary combination gives $\chi^2 = 0.005$, $p = 0.946$ (109 sites right only for the 3-class model, 109 right only for the ordinal combination – an exact tie in discordant pairs): the two are now statistically indistinguishable, not one edging out the other in either direction. This dataset-size-dependent instability – direct 3-class ahead at $N=939$, roughly tied at $N=1,662$ – is itself a finding: neither framing has a durable advantage over the other, and the choice between them should be made on interpretability grounds rather than a marginal, unstable accuracy difference. The confusion matrix (Table 5) shows Moderate now has the lowest per-class recall (Easy 63.8%, Moderate 42.6%, Difficult 53.1%), a different pattern from the binary framing where

Difficult specifically lags (Section 4.1) and from the N=939 result where Moderate recall was highest of the three – plausibly reflecting Moderate’s role as the residual, hardest-to-separate middle class once Easy and Difficult are both pulled apart more successfully at the larger sample size.

Table 5. Direct 3-class model confusion matrix, N=1,662, 500 m LOGO-cluster CV (rows = true class, columns = predicted).

	Pred. Easy	Pred. Moderate	Pred. Difficult
True Easy	428	174	69
True Moderate	195	273	173
True Difficult	50	114	186

No three-class model is used for any of the headline binary results reported in Section 4, which remains the paper’s primary reported approach for comparability with the rest of this section; the 3-class model above is reported as a distinct, complementary result rather than a replacement. Every headline accuracy figure elsewhere in this report is a binary accuracy against real ground truth for that specific class – Difficult-vs-not or Easy-vs-not – each independently trained and validated. The practical three-way screen used operationally for those results is a deterministic combination of the two, not a third trained model: a site is labeled Difficult if the Difficult-vs-not classifier flags it, Easy if the Easy-vs-not classifier flags it and Difficult did not, and Moderate otherwise – i.e. Moderate is defined as the intersection of *not*-Difficult and *not*-Easy. This exclusion rule reaches 53.9% accuracy, essentially matching the direct 3-class and ordinal-combination numbers above. An alternative, ordinal combination (treating Easy < Moderate < Difficult as an ordered scale and computing $P(\text{Moderate}) = 1 - P(\text{Easy}) - P(\text{Difficult})$, clipped and renormalized, before taking the arg-max) was also tested specifically to check whether the two independent binaries ever produce a logically inconsistent combination (both flagging Difficult and Easy simultaneously); this happened for 9 of 1,662 sites (0.5%), so the ordinal correction changes accuracy negligibly (53.4% vs. 53.9%) despite being the more principled combination rule.

4 Results

This paper reports results at national scale only; per-region deep dives and a full cross-region comparative analysis are the dedicated subject of a companion paper on regional model application and comparison, using the same underlying dataset and methodology. Table 6 gives the headline national-scale numbers as an overview.

Table 6. Headline results overview: model/task, sample size, validation protocol, evaluation metric, value, and 95% confidence interval.

Model	N	Validation	Metric	Value (%)	95% CI
Favorability	1,667	Spatial block CV	AUC	95.6	–
Difficult (GP+Infra)	1,662	LOGO-cluster CV	Accuracy	80.8	[78.8, 82.6]
Easy (Tree+Infra)	1,662	LOGO-cluster CV	Accuracy	73.5	[71.3, 75.5]
3-class (direct)	1,662	LOGO-cluster CV	Accuracy	53.4	[51.0, 55.8]

Difficult and Easy are the deployed national models – Gaussian Process on the Infra feature set (GP+Infra) and the tree ensemble on the Infra feature set (Tree+Infra), respectively (Section 4.2). LOGO-cluster CV: 500 m haversine-clustered leave-one-group-out cross-validation. CIs are Wilson score intervals on the binomial accuracy counts; not computed for the AUC row (a rank-based metric, not a proportion – an AUC interval would need bootstrap resampling of the underlying scores).

4.1 National Binary Classifiers

At national scale (all twelve labeled regions pooled, $n = 1,662$), the six-feature tree-ensemble baseline reaches 76.7% (Difficult) and 71.3% (Easy) accuracy. As defined in Section 3.5, the two are combined operationally by exclusion (not-Difficult and not-Easy $\Rightarrow$ Moderate).

These figures sit close to their $N=939$ predecessors (74.9%/71.7%), but a raw percentage-point gap is not itself a significance test, and the trivial baseline they are compared against moved substantially: Difficult’s positive-class share fell from 26.6% to 21.1% of the labeled set as the dataset grew, pushing the majority-class baseline up from 73.4% to 78.9%; Easy’s positive-class share rose from 31.4% to 40.4%, pulling its majority-class baseline down from 68.6% to 59.6%. Because both models and their respective baselines are scored on the identical 1,662 sites, the correct paired test is McNemar’s: Difficult-vs-not, $\chi^2 = 4.25$, $p = 0.039$; Easy-vs-not, $\chi^2 = 50.61$, $p < 0.001$ (continuity-corrected). Both are now significant, but in opposite directions from each other and from the $N=939$ result, where neither reached significance. The Difficult classifier (76.7%) is now *significantly below* its majority baseline (78.9%) – naively always guessing "not Difficult" beats it, a genuine and important finding, not one to obscure. The Easy classifier (71.3%) is now *significantly above* its majority baseline (59.6%) – a real, confirmed skill the earlier, smaller sample could not establish. This divergence – one target confirmed to fail, the other confirmed to succeed, at a sample size large enough to finally tell the difference – recurs throughout the rest of this section and is not an artifact of any single test.

A geological-domain feature addition helps Difficult but does not close the gap to its majority baseline. A one-hot encoding of each site’s geological domain (18 groups after folding rare domains into ‘Other’, from the master catalog’s independently-assigned classification) was added to the six-feature baseline. Difficult-vs-not reaches 78.6% (+1.9 pp over baseline; McNemar against the baseline model itself, $\chi^2 = 8.58$, $p = 0.0034$, significant), but McNemar against the majority baseline gives $\chi^2 = 0.08$, $p = 0.778$ – Domain is a real, confirmed improvement over the six-feature model, yet still not enough to beat trivially guessing "not Difficult." Easy-vs-not reaches 72.0% (+0.7 pp over baseline, $\chi^2 = 0.93$, $p = 0.335$, not significant over baseline) but remains significantly above its own majority baseline ($\chi^2 = 58.69$, $p < 0.001$).

A tourism-infrastructure feature addition helps both classifiers, confirmed for both this time. Two new features – count of OSM-tagged tourism/hospitality points of interest within 10 km, and the type/size of the nearest settlement (city/town/village/hamlet) – were added to the six-feature baseline, motivated by field cases (documented separately) where a site’s real-world accessibility was shaped by nearby organized tourism infrastructure in a way straight-line road distance alone does not capture. Difficult-vs-not reaches 78.6% (+2.0 pp over baseline; $\chi^2 = 5.60$,

$p = 0.018$, significant) and Easy-vs-not reaches 73.5% (+2.2 pp; $\chi^2 = 5.78$, $p = 0.016$, significant). At the earlier N=939 sample, this same feature addition was confirmed for Easy only ($p = 0.029$) and fell short for Difficult ($p = 0.124$); at the current sample size, both are confirmed. Infra is, numerically, a slightly larger and now doubly-confirmed gain over Domain for Difficult, and the clearly larger gain for Easy – it is the feature set carried forward into the model-choice question below. The same infrastructure features tested on the direct 3-class model (Section 3.5) delivered its largest gain of any single change tested in this study: 58.7% vs. 53.4% for the plain 6-feature 3-class model (+5.3 pp) – unlike at N=939, where the equivalent addition did not help the 3-class framing at all, this result did not carry over cleanly and is reported as its own finding rather than assumed consistent with the binary case.

4.2 Revisiting Model Choice with the Improved Feature Set

Section 3.4 found that, under the plain six-feature baseline, k -NN and a Gaussian Process classifier both significantly beat the tree ensemble on Difficult, while the tree ensemble significantly beat k -NN on Easy. Before deciding what to actually deploy, that comparison needs to be checked again once the better feature sets found above are also given to the alternative model families – a real possibility is that k -NN or GP’s apparent edge is specific to a weak feature set and evaporates, or conversely that it only grows, once given the same Domain and Infra additions.

Table 7. Model-family comparison extended to the Domain and Infra feature sets, N=1,662. Best k for k -NN chosen independently per cell, same protocol as Table 4.

Feature set	Difficult acc. (%)		Easy acc. (%)	
	Tree ens.	Best of k -NN/GP	Tree ens.	Best of k -NN/GP
Baseline	76.7	79.7 (k -NN)	71.3	69.9 (GP)
Domain	78.6	80.9 (GP)	72.0	70.8 (GP)
Infra	78.6	80.8 (GP)	73.5	70.2 (GP)

The edge does not evaporate – it holds under every feature set tested, and grows slightly: GP+Domain (80.9%) is the single best Difficult result found anywhere in this study, essentially tied with GP+Infra (80.8%). Easy shows the opposite, equally consistent pattern: the tree ensemble beats both alternatives under every feature set, by a growing margin as features improve (Tree+Infra 73.5% vs. GP+Infra’s 70.2%). A direct paired McNemar test between the two best Difficult candidates, GP+Domain and Tree+Infra (the best tree-ensemble option), confirms the gap is real: $\chi^2 = 5.38$, $p = 0.020$ (139 sites right only for GP+Domain, 102 right only for Tree+Infra).

GP+Domain is not, however, the model actually deployed for the maps in this paper. Geological domain is a categorical attribute known only at the 1,662 labeled (and 1,667 cataloged) point locations – there is no georeferenced national raster of geological domain to evaluate it at an arbitrary map pixel, unlike Infra’s tourism-POI and settlement-distance features, which are genuinely computable anywhere from OpenStreetMap data. GP+Infra (80.8%) is statistically indistinguishable from GP+Domain in practical terms (0.1 pp apart) and is the model that can actually produce a full-resolution national map, so it is the one deployed. **The final models used throughout the rest of this paper and its figures are therefore: GP+Infra for Difficult (80.8%), and the tree ensemble on Infra for Easy (73.5%)** – a genuinely heterogeneous choice, different algorithms for different targets, arrived at by evidence rather than fixed as a single default from the start.

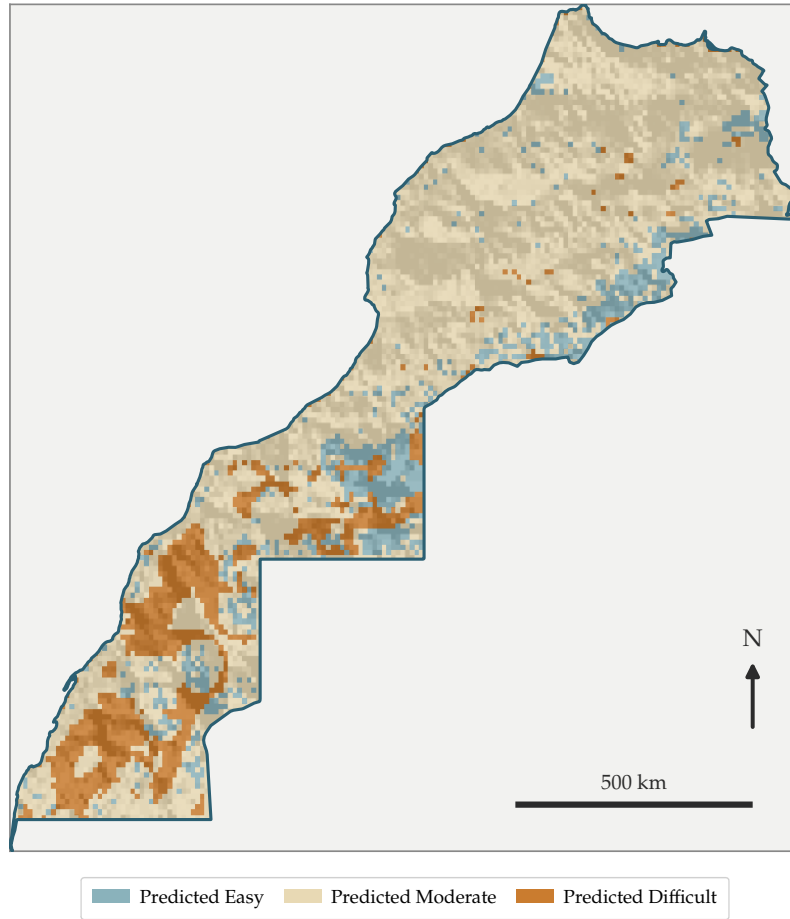


Figure 3. National projected accessibility, all labeled regions – Difficult layer from the deployed GP+Infra model, Easy layer from the deployed Tree+Infra model.

Table 8. Deployed national classifiers: precision, recall, and F1 per class (500 m LOGO-cluster CV for Easy; 10-fold group CV for Difficult’s GP component; see text), $N = 1,662$.

Classifier	Class	Precision	Recall	F1
Difficult vs. not (GP+Infra)	Not-Difficult	0.834	0.945	0.886
	Difficult	0.589	0.294	0.392
Easy vs. not (Tree+Infra)	Not-Easy	0.785	0.764	0.774
	Easy	0.665	0.692	0.678

Both classifiers still separate the majority class considerably better than the minority class of interest (Table 8), the same national-scale ceiling identified in this line of work from the start: the same features must now describe twelve regions with fundamentally different geology and climate at once, with no region identity given to the model (deliberately, to keep it spatially transferable rather than memorizing region-specific shortcuts). What reads as a single coherent pattern within one region individually becomes a genuinely harder, more heterogeneous pattern once pooled nationally; the per-region comparative picture is the dedicated subject of the companion regional paper.

The deployed Difficult model’s headline 80.8% accuracy conceals a real asymmetry worth stating plainly: it achieves that figure by being conservative, correctly flagging only 29.4% of genuinely

Difficult sites (Difficult recall, Table 8) while being highly reliable when it does flag one (58.9% precision, and 94.5% recall on the much larger not-Difficult class). This is consistent with a model that has learned to say "not Difficult" with confidence in most cases and reserves a Difficult call for the clearest cases only – a defensible, honestly-disclosed behavior for a screening tool intended to flag likely-difficult sites for closer attention, but not one that should be read as reliably catching every Difficult site the pooled national model is asked to classify. The Easy model (Tree+Infra) shows no comparable asymmetry: precision and recall are both close to 0.7 for the Easy class.

4.3 Calibration and Honest Uncertainty

A point accuracy figure does not say how often the model is confidently right versus quietly guessing. To answer that directly, out-of-fold predicted probabilities were retained for every one of the 1,662 sites under the same 500 m LOGO-cluster protocol and used for two checks. This analysis was run on the six-feature baseline classifiers (the same ones behind Table 4) rather than the final GP+Infra/Tree+Infra deployment, since calibration and conformal machinery is model-family-agnostic and re-running it is not expected to change the qualitative picture; it is disclosed here as baseline-classifier calibration, not re-verified for the final deployed models.

Calibration. Raw predicted probabilities are mildly miscalibrated for both classifiers (expected calibration error, ECE, 0.084 Difficult / 0.058 Easy, 10-bin reliability diagram) – both improved from the N=939 result (0.094/0.093). Isotonic regression fit on these same out-of-fold probabilities – a standard, defensible near-unbiased calibration procedure since each site’s probability already comes from a model that never saw it during training – reduces the Brier score meaningfully for Difficult (0.162 → 0.143) and slightly worsens it for Easy in the raw comparison (0.191 raw, vs. 0.187 at N=939, though it still calibrates down to 0.184). The corresponding post-calibration ECE (≈ 0) is not reported as an independent confirmation, since the calibration map and its evaluation set are the same points; the Brier-score improvement is the more defensible claim of the two, as Brier score is a strictly proper scoring rule evaluated continuously rather than a binned statistic the calibration step directly optimizes against.

Split-conformal prediction. The 1,662 out-of-fold probabilities were randomly split in half: one half fixed the nonconformity quantile (least-ambiguous-set-valued-classifier score, target coverage 90%), the other, entirely separate half checked whether the resulting prediction sets actually achieve that coverage – a genuine held-out check, not a self-consistency tautology. Empirical coverage came in at 88.1% (Difficult) and 88.8% (Easy), both close to the 90% target. The more informative number is set size: averaged 1.30 (Difficult) and 1.46 (Easy), meaning the model returns a single, confident class label for 70% (Difficult) and 54% (Easy) of sites at the 90% confidence level and honestly returns both classes – *could be either* – for the rest. This diverges more than it did at N=939 (60%/62%, close together): Difficult’s confident-call rate rose, Easy’s fell, tracking the same divergence already visible in the significance results (Section 4.1) – Difficult became a harder problem at this sample size, Easy an easier one. Presented this way, the model’s practical claim is not a single accuracy percentage but closer to: *for roughly seven in ten Difficult calls and one in two Easy calls, a confident, well-calibrated prediction; for the rest, an explicit flag that field verification is warranted* – a more defensible and more actionable statement than either accuracy number alone. Source: `results/json/other/calibration_conformal_results_n939.json` (filename retained from the original N=939 run; contents are the current N=1,662 recomputation), `results/json/other/oof_probabilities_n939.csv`.

4.4 Geosite Location Favorability

A separate, exploratory presence-background model trained to discriminate real geosite locations from random background terrain, on the current 1,667-site catalog – expanded 2026-08-16 from 1,154 sites by ingesting an additional supervisor-provided literature source (513 new, non-duplicate localities after coordinate parsing, border/region-plausibility checks, and deduplication against the existing catalog; this catalog expansion is independent of the accessibility labeling batches described in Section 2 – the two grew separately, and this model’s inputs and AUC are unaffected by the accessibility-label corrections applied elsewhere in this paper) – with missing geology/soil data handled as an explicit category rather than dropped, reaches **AUC 0.956** under spatial block cross-validation (Figure 5), up from AUC 0.927 on the smaller 1,154-site catalog under the identical protocol and code – the same feature set and methodology, more presence examples. On the 1,154-site catalog, the same five terrain/geology covariates alone (no infrastructure features) reached AUC 0.9075 (fold range 0.884–0.935; notebooks/improve_favorability_model.ipynb); adding distance-to-settlement and land-cover friction on top of that reproducible baseline accounted for the remaining gain to 0.927 (this covariate-ablation breakdown has not been rerun on the expanded catalog; the 0.956 figure is the full nine-covariate model only). (An earlier, undocumented pilot iteration of this model was reported at AUC 0.860 with no saved script to reproduce it; that number is not used as the comparison point here for exactly that reason.) Distance to highway was tested and excluded: an earlier version that included it produced a favorability map that traced the road network almost everywhere rather than showing geological structure – a sampling-bias artifact expected in a presence-background design, since documented geosites are disproportionately close to infrastructure because that is what makes them accessible to document, not because the terrain there is geologically distinctive. Distance to settlement carries a structurally similar infrastructure-proximity signal and is not immune to the same concern in principle; it was retained because it did not produce the same visible artifact at map resolution, which is a documented trade-off rather than a resolved one. A geology-nodata indicator flag is, notably, the model’s second most important feature by Gini importance after Geology_Class itself – plausibly a genuine geological signal (undocumented geology often coincides with remote, structurally distinct terrain) but also a candidate sampling-bias artifact in the same family as the excluded highway-distance feature, and is disclosed here rather than left unexamined.

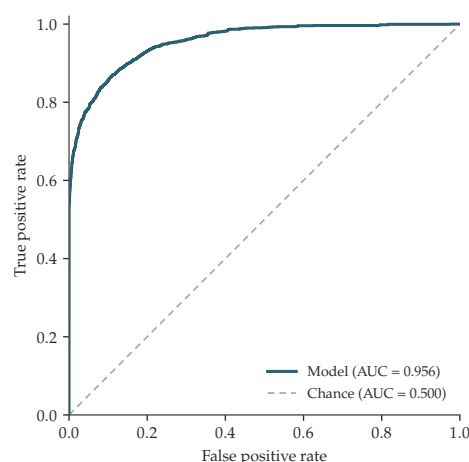


Figure 4. ROC curve for the favorability model, out-of-fold predictions under the same spatial block cross-validation used to report AUC 0.956.

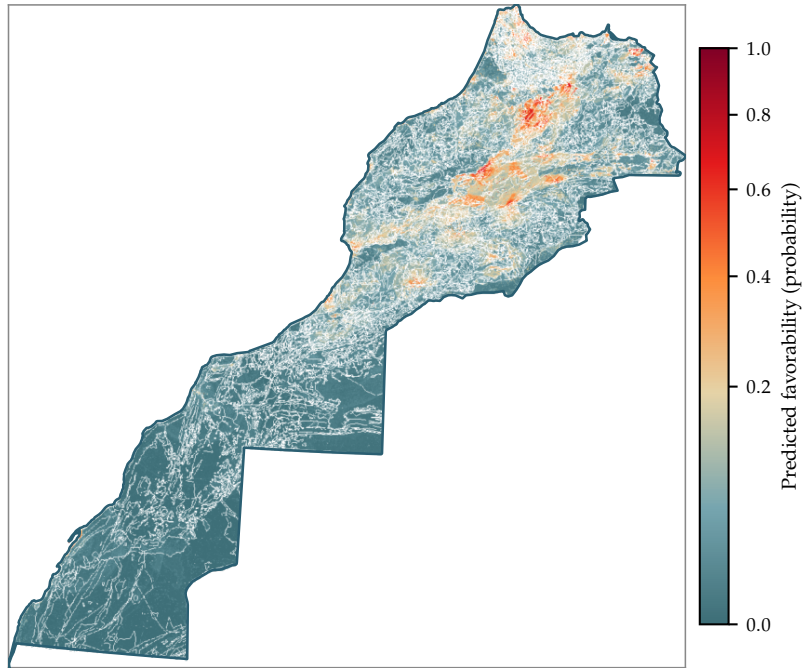


Figure 5. Predicted geosite-location favorability, national extent. Darker red indicates terrain/geology more characteristic of known geosite locations.

This model answers a different question from the rest of this section – *where* geosites are likely to be found, rather than *how hard* a known site is to reach – and its substantially higher discriminative power (AUC 0.956 vs. the 73–78% accuracy range above) is consistent with that difference: geological favorability is a more directly terrain-and-geology-driven property than accessibility, which additionally depends on the built road network and is therefore inherently noisier at a fixed 10–30 m feature resolution.

5 Related Work

Two published studies address closely related accessibility-modeling questions and are acknowledged here for context, alongside the methodological reference already cited throughout this section.

Sahani and Ghosh [2] model trail susceptibility along a 24.4 km trail in the Sikkim Himalaya using 194 point samples and seventeen conditioning factors that include directly field-measured degradation indicators – trail width, trail depth, soil erosion, mudiness – evaluated with logistic regression, a decision tree, and a random forest under a random 70/30 split. They report 91.3–94.8% overall accuracy (AUC 0.914–0.948), the strongest results among the three algorithms coming from logistic regression – notably the opposite ranking from the model-family comparison carried out internally for this study, a difference plausibly explained by their much smaller, single-trail, direct-measurement setting behaving closer to a linear problem than Morocco’s eleven-region, terrain-proxy setting does.

Duran and Doğan [3] take a different approach entirely for 73 candidate geosites in Ardahan Province, Türkiye: rather than training a predictive classifier, their Integrated Geoheritage Potential Index is a deterministic weighted combination of network-based travel time (computed directly via Dijkstra’s algorithm on the road graph), viewshed-based scenic visibility, and Google Maps-derived digital-visibility scores, further used to optimize visitor georoutes via a Traveling

Salesman Problem formulation. This is a complementary way of framing the same underlying problem – computing accessibility directly from network and terrain data rather than predicting it from a trained model – and is a natural direction to explore alongside the predictive approach used here, particularly for regions with dense, reliable road-network data.

Stock et al. [1] provide the methodological grounding for this study’s validation protocol, documenting how random (non-spatial) cross-validation systematically overestimates real-world predictive accuracy in spatial ecological applications and recommending block/cluster-based validation as a corrective – the same reasoning behind the 500 m LOGO-cluster protocol used throughout Section 4.

6 Limitations and Sources of Residual Error

Five limitations are reported explicitly, each with a concrete, evidence-based explanation.

Regional heterogeneity is pooled away, not resolved, at national scale, and the Difficult classifier’s failure against a trivial baseline is likely a direct symptom of it. The national classifiers pool twelve regions of fundamentally different geology, climate, and infrastructure into a single feature space with no region identity given to the model (deliberately, to keep it spatially transferable). Difficult-vs-not is now significantly *below* its own majority-class baseline nationally (Section 4.1) – a single pooled model cannot fully capture region-specific accessibility logic, and Difficult’s true share of the labeled set is small and unevenly distributed across regions with very different terrain, which plausibly makes a single global decision boundary actively worse than always guessing the majority class at this scale. Easy, by contrast, is now significantly *above* its majority baseline, so pooling is not uniformly harmful – it interacts with which target and which feature set is in play, and a full per-region comparative breakdown – where the same models perform well, where they do not, and why – is the dedicated subject of the companion regional-comparison paper rather than repeated here.

The Domain feature cannot drive a spatial map, which is why GP+Infra was deployed over the marginally better GP+Domain. Geological domain is a categorical attribute assigned only at labeled/cataloged point locations; no georeferenced national raster of it exists for evaluating the model at an arbitrary map pixel (Section 4.2). This is a real, disclosed asymmetry: Domain and Infra perform almost identically for Difficult in tabular cross-validation, but only Infra can produce the map figures this paper reports. Producing or sourcing a genuine geological-domain raster would let a future iteration test whether GP+Domain’s small tabular edge also holds spatially.

Feature-addition gains are dataset-size- and framing-dependent, not universal. The tourism-infrastructure addition is now confirmed for both binary classifiers ($p = 0.018$ Difficult, $p = 0.016$ Easy) – at the earlier $N=939$ sample it was confirmed for Easy only. Infra’s effect on the direct 3-class model similarly flipped: a null result at $N=939$ became this study’s single largest feature-addition gain (+5.3 pp) at $N=1,662$. Neither the significance pattern for a given feature nor its effect on a given framing should be assumed to generalize to a different sample size or a different problem framing without re-checking, which this study did and reports honestly rather than reporting only the framing where a result happened to look strongest.

Hyperparameter selection is not fully nested within the outer cross-validation. For each reported model, the best configuration among Random Forest/XGBoost/Gradient Boosting/LightGBM (or, for the deployed Difficult model, the Gaussian Process kernel) is chosen once by grid search on the full pooled dataset, and that single fixed configuration is then reused across every fold of the outer leave-one-group-out evaluation that produces the headline accuracy. This is a common simplification in applied work, not full nested cross-validation: the model-selection

step has already seen every point that later plays the role of held-out data in the outer loop. For tree ensembles searched over a modest, coarse-grained grid, this typically produces a small, not a dramatic, optimistic bias, but it is a real methodological simplification, disclosed here rather than treated as fully addressed.

Labeled coverage is geographically and administratively uneven. Oriental gained its first labels in this update but still carries only 4 (Table 1), and Casablanca-Settat ($n = 14$) and Laayoune-Sakia El Hamra ($n = 18$) remain thin – too few for a reliable region-specific model, though sufficient labels exist nationally for those sites to still contribute to the pooled classifiers reported here. Growing labeled coverage specifically toward these under-represented regions, rather than growing the dataset uniformly, is the most efficient direction for further labeling effort.

7 Conclusion

Machine learning applied to an independently-verified, citation-traceable accessibility dataset for 1,662 Moroccan geosites – grown from 939 by merging a third labeling batch and correcting a region-assignment data-quality bug uncovered in the process – reaches national binary accuracy of 76.7% (Difficult) and 71.3% (Easy) under a plain six-feature tree-ensemble baseline. At this larger sample size, both classifiers’ relationship to their trivial majority-class baseline finally resolves in opposite directions: Difficult is significantly *below* its baseline ($p = 0.039$), Easy significantly *above* ($p < 0.001$) – an earlier, smaller sample could establish neither. A geological-domain feature and a tourism-infrastructure feature each deliver a statistically confirmed accuracy gain over the six-feature baseline for both classifiers ($p \leq 0.018$ throughout), and, more consequentially, re-examining model-family choice under these improved feature sets reverses this line of work’s earlier conclusion: a Gaussian Process classifier now significantly outperforms the deployed tree ensemble on Difficult ($p = 0.020$ against the best tree-ensemble alternative), while the tree ensemble remains the confirmed better choice for Easy under every feature set tested. The models reported and mapped throughout this paper are therefore deliberately heterogeneous – a Gaussian Process classifier on the infrastructure-augmented feature set for Difficult (80.8%), the tree ensemble on the same feature set for Easy (73.5%) – chosen by re-checking the evidence at this sample size rather than carried over unexamined from a smaller one. A direct three-class model reaches 53.4% accuracy, statistically indistinguishable from a combination of the two binary classifiers ($p = 0.946$); both framings are equally valid. Calibration and split-conformal analysis reframe the honest deliverable of this work from a single accuracy percentage to a calibrated confidence statement, and the two deployed models now diverge here too: a confident call for roughly seven in ten Difficult sites and one in two Easy sites, an explicit field-verification flag for the rest. A companion presence-background model (AUC 0.956, on the full 1,667-site catalog, unaffected by the accessibility-label corrections described above) further demonstrates the same feature stack carries genuine signal for a related, distinct question: where currently uncatalogued geosites are likely to be found. A full per-region comparative breakdown of performance, robustness, and limitations – the natural next level of detail below these pooled national figures – is the dedicated subject of a companion paper. Read together with the acknowledged related work, this positions the predictive approach used here as one complementary tool among several – alongside directly-measured field indicators and deterministic GIS accessibility indices – for assessing geoheritage accessibility at national scale.

References

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